

No. of Functional Features Included in the Solution

Date	7 July 2026
Team ID	XXXXXX
Project Name	Smart Lender
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status	Marks Contribution
1	Loan Application Form	User fills 11 fields (gender, income, loan amount, credit history, etc.) and submits for prediction	Frontend – index.html	Done	High
2	ML Model Training	Trains 4 models (Decision Tree, Random Forest, KNN, XGBoost), compares accuracy, selects best	Backend – train_model.py	Done	High
3	Loan Eligibility Prediction	Predicts Approved or Rejected based on user inputs using saved XGBoost model	Backend – app.py (predict())	Done	High
4	Prediction Result Page	Displays Approved/Rejected result with a full applicant summary table	Frontend – result.html	Done	Medium
5	Key Decision Factors Display	Shows 4 factors (credit history, income, property area, education) that influenced the decision	Frontend – result.html	Done	Medium
6	Data Preprocessing Pipeline	Handles missing values, encodes categorical features, splits data for training	Backend – train_model.py	Done	Medium
7	Model Persistence (Pickle)	Saves trained XGBoost model as model.pkl and loads it at app startup without retraining	Backend – app.py / model.pkl	Done	Low
8	Responsive UI with Stats	Navbar, hero section showing 94.7% train / 81.1% test accuracy, How It Works section, footer	Frontend – style.css / index.html	Done	Low

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	5 (Form, Model Training, Prediction, Result Page, Preprocessing)
Additional / Nice-to-Have Features	3 (Decision Factors, Model Persistence, Responsive UI with Stats)
Features Tested & Verified	8

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	3	Loan Application Form, Prediction Result Page, Responsive UI with Stats & How It Works section
2	Backend / Logic	4	ML Model Training, Loan Eligibility Prediction, Data Preprocessing Pipeline, Model Persistence
3	Database / Storage	1	CSV dataset (loan.csv) used for training; model.pkl for persistent model storage
4	API / Integration	1	Flask REST route /predict (POST) connects frontend form to backend ML model
5	Security / Authentication	1	Input validation on all form fields; try-except error handling prevents crashes on bad input